Analytical MCQ Assessment: Function and Array

High-level cognitive evaluation paper.

SECTION A: Multiple Choice Questions

Q1. What is the primary purpose of using the 'map' function in an array? (1 Marks)

A. To filter out specific elements from the array
B. To transform each element in the array
C. To sort the elements in the array
D. To reverse the order of elements in the array

Q2. Which of the following array methods returns a new array with all elements that pass the test implemented by the provided function? (1 Marks)

A. forEach
B. map
C. filter
D. reduce

Q3. What is the difference between 'null' and 'undefined' in JavaScript? (1 Marks)

A. 'null' represents the absence of any object value, while 'undefined' represents an uninitialized variable
B. 'null' represents an uninitialized variable, while 'undefined' represents the absence of any object value
C. 'null' and 'undefined' are interchangeable terms
D. 'null' is a string, while 'undefined' is a number

Q4. Which of the following functions is used to invoke a function with a given 'this' value and arguments? (1 Marks)

A. call
B. apply
C. bind
D. All of the above

Q5. What is the purpose of the 'hasOwnProperty' method in JavaScript? (1 Marks)

A. To check if an object has a property
B. To check if an object has a method
C. To check if an object has a prototype
D. To check if an object has a constructor

Q6. Which of the following is an example of a higher-order function? (1 Marks)

A. A function that takes a string as an argument
B. A function that returns a number
C. A function that takes another function as an argument

D. A function that returns an object

Q7. What is the purpose of the 'slice' method in an array? (1 Marks)

A. To remove elements from the array
B. To add elements to the array
C. To extract a section of the array
D. To sort the array

Q8. Which of the following is an example of a closure? (1 Marks)

A. A function that has access to its own scope
B. A function that has access to the global scope
C. A function that has access to the scope of its outer function
D. A function that has no access to any scope

Q9. What is the purpose of the 'every' method in an array? (1 Marks)

A. To check if all elements in the array pass a test
B. To check if any element in the array passes a test
C. To check if the array is empty
D. To check if the array is not empty

Q10. Which of the following is an example of a recursive function? (1 Marks)

A. A function that calls another function
B. A function that calls itself
C. A function that returns a value
D. A function that takes an argument

INSTRUCTOR ONLY - ANSWER KEY & RUBRIC

MCQ Solutions & Rationale

Q1. Correct Answer: B

Rationale: The 'map' function is used to create a new array with the results of applying a provided function on every element in the calling array. This is why option B is correct, as it accurately describes the primary purpose of the 'map' function. Options A, C, and D are incorrect because they describe the purposes of the 'filter', 'sort', and 'reverse' functions, respectively.

Q2. Correct Answer: C

Rationale: The 'filter' method creates a new array with all elements that pass the test implemented by the provided function. This is why option C is correct. Options A, B, and D are incorrect because 'forEach' is used to execute a function for each element, 'map' is used to transform elements, and 'reduce' is used to reduce the array to a single value.

Q3. Correct Answer: A

Rationale: In JavaScript, 'null' represents the absence of any object value, while 'undefined' represents an uninitialized variable or a variable that has not been declared. This is why option A is correct. Options B, C, and D are incorrect because they either reverse the meanings of 'null' and 'undefined' or assign incorrect data types to them.

Q4. Correct Answer: D

Rationale: The 'call', 'apply', and 'bind' functions are all used to invoke a function with a given 'this' value and arguments. The 'call' function invokes the function immediately, the 'apply' function invokes the function immediately with an array of arguments, and the 'bind' function returns a new function that can be invoked later. This is why option D is correct. Options A, B, and C are incorrect because they only partially describe the correct answer.

Q5. Correct Answer: A

Rationale: The 'hasOwnProperty' method returns a boolean indicating whether an object has a property as its own property (as opposed to inheriting it). This is why option A is correct. Options B, C, and D are incorrect because they describe different purposes, such as checking for methods, prototypes, or constructors.

Q6. Correct Answer: C

Rationale: A higher-order function is a function that takes another function as an argument or returns a function as a result. This is why option C is correct, as it describes a function that takes another function as an argument. Options A, B, and D are incorrect because they describe functions that take or return different types of data, but do not necessarily involve functions as arguments or return values.

Q7. Correct Answer: C

Rationale: The 'slice' method returns a shallow copy of a portion of an array. This is why option C is correct, as it describes the purpose of extracting a section of the array. Options A, B, and D are incorrect because they describe different methods, such as 'splice' for removing elements, 'push' or 'unshift' for adding elements, or 'sort' for sorting the array.

Q8. Correct Answer: C

Rationale: A closure is a function that has access to the scope of its outer function, even when the outer function has returned. This is why option C is correct, as it describes a function that has access to the scope of its outer function. Options A, B, and D are incorrect because they describe different scenarios, such as a function with access to its own scope, the global scope, or no scope at all.

Q9. Correct Answer: A

Rationale: The 'every' method tests whether all elements in the array pass the test implemented by the provided function. This is why option A is correct, as it describes the purpose of checking if all elements pass a test. Options B, C, and D are incorrect because they describe different methods, such as 'some' for checking if any element passes a test, or scenarios related to the emptiness of the array.

Q10. Correct Answer: B

Rationale: A recursive function is a function that calls itself, either directly or indirectly. This is why option B is correct, as it describes a function that calls itself. Options A, C, and D are incorrect because they describe different scenarios, such as a function that calls another function, returns a value, or takes an argument, but do not necessarily involve self-reference.